System Identification & Compensator Design of the Propeller-Powered Racing Vehicle

Course: Practice of Mechanical Engineering (Group 12)
The Experiment was designed and discussed by Tsung-Wun Wang (王琮文).

The objective of the experiment is to identify the system of our propeller-powered racing vehicle, and to design a controller that can improve the performance from root locus analysis.

A. The Propeller-Powered Racing Vehicle

In this capstone course, we are required to make our own propeller-powered racing vehicle. You can see our vehicle in Figure 1.

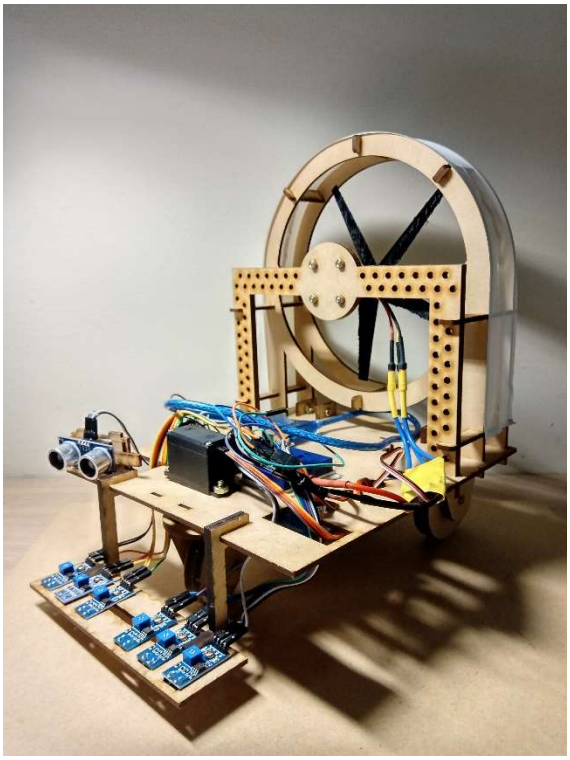

Figure 1. The Propeller-Powered Racing Vehicle

B. System Identification

1. Data retrieval

First of all, many different patterns of encoder were designed. Although the complicated pattern could improve the accuracy, we choose the encoder shown in Figure 3 because of the limit of the infrared sensor.

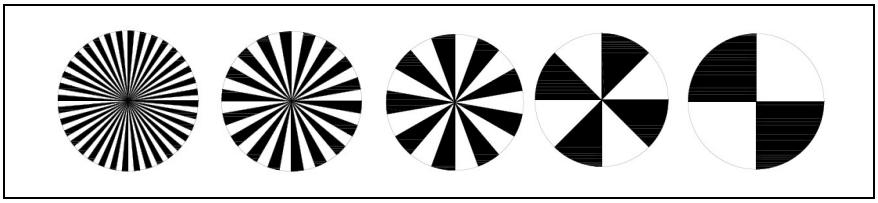

Figure 2. Patterns of Encoders

Then, we installed the infrared sensor on one of the rear wheels.

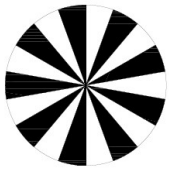

Figure 3. Our Encoder.

Figure 4. Installation of the Infrared Sensor.

We get the pulse-width modulation (PWM) and revolutions per minute (rpm) of the propeller.

2. Data Pre-processing

(1) Fixed PWM (PWM=30)

The tangential velocity can be calculated from the rpm of the propeller. To be more specific, the diameter of the wheel is 65 mm, and circumference is $65 \times \pi/18$ (mm). The experiment is repeated five times, and the following results is from the fourth experiment.

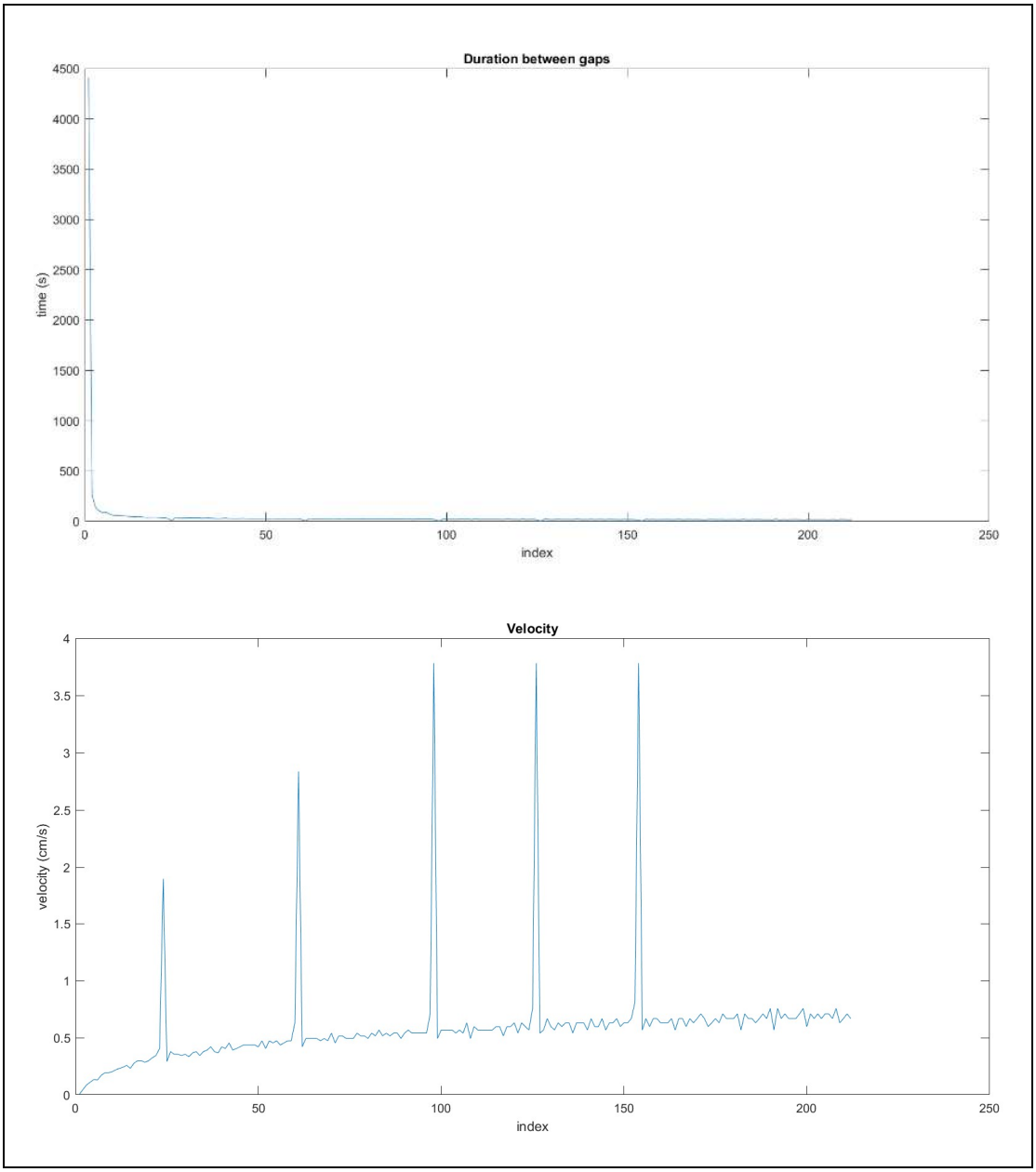

Figure 5. The tangential velocity of the propeller calculated from the fourth experiment (Fixed PWM).

From Figure 5, the result is reasonable from the beginning except for few noises.

(2) Random PWM of the propeller: random inputs are in the range of [20,50]

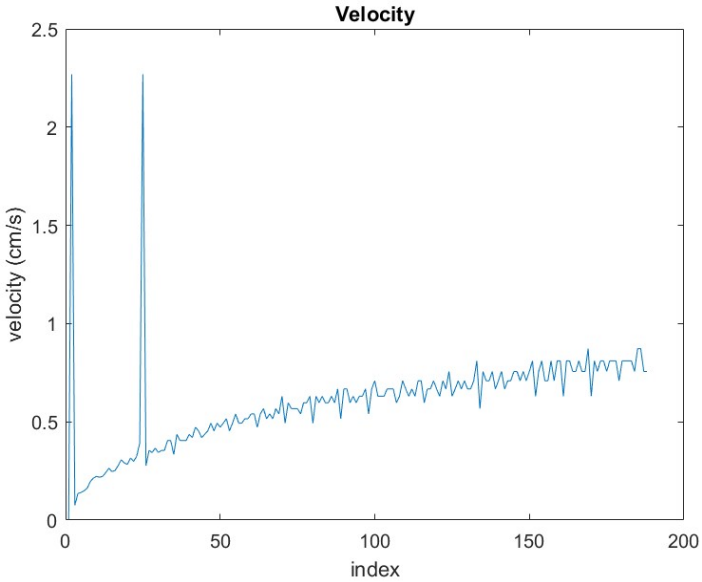

Figure 6. The tangential velocity of the propeller calculated from the fourth experiment (random PWM).

3. System Identification

We refered to matlab 'systemIdentification' module to finish the system identification [1].

input: PWM of the propeller.

output: linear velocity of the vehicle (cm/s).

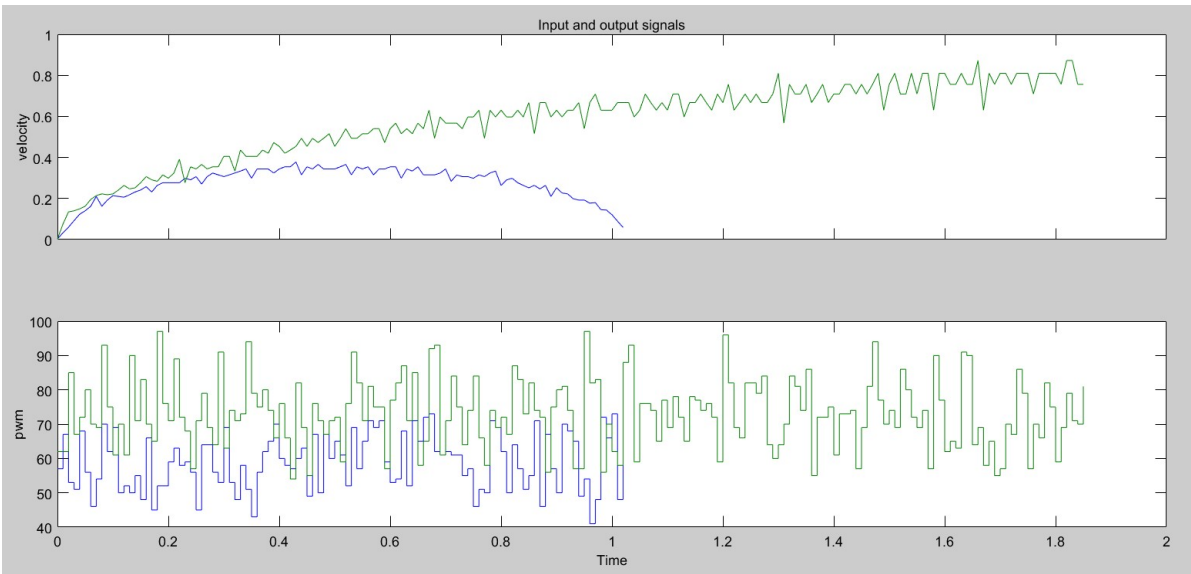

Figure 7. Comparison between the 2nd and the 4th experiment.

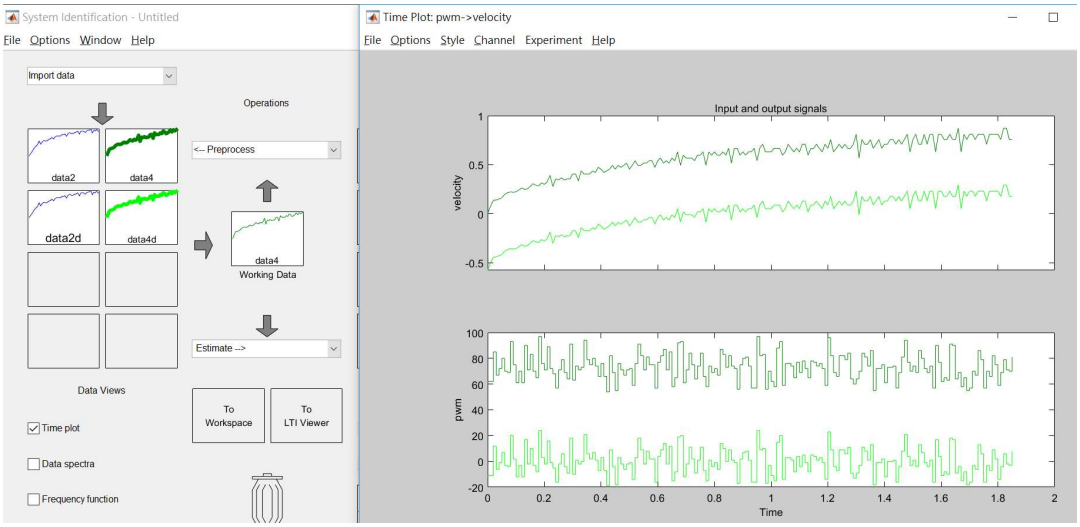

Figure 8. Output before and after subtracting the noise (4th experiment).

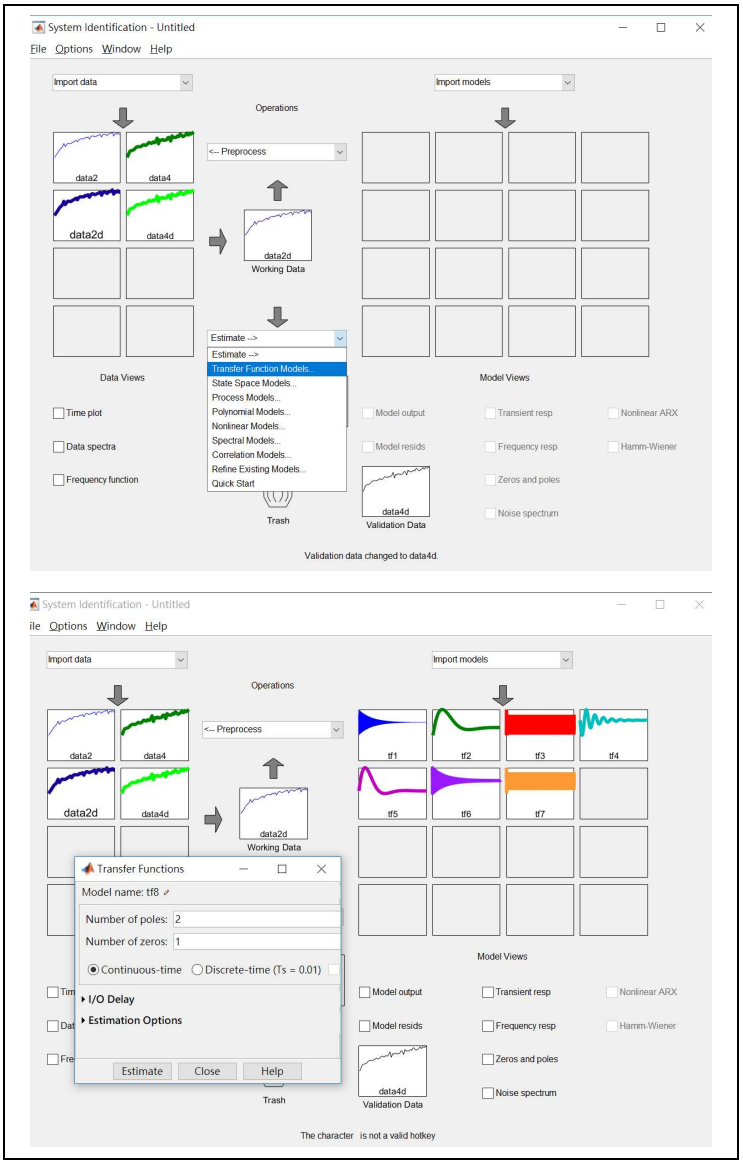

Figure 9. The choice of the number of poles and zeros of the transfer function.

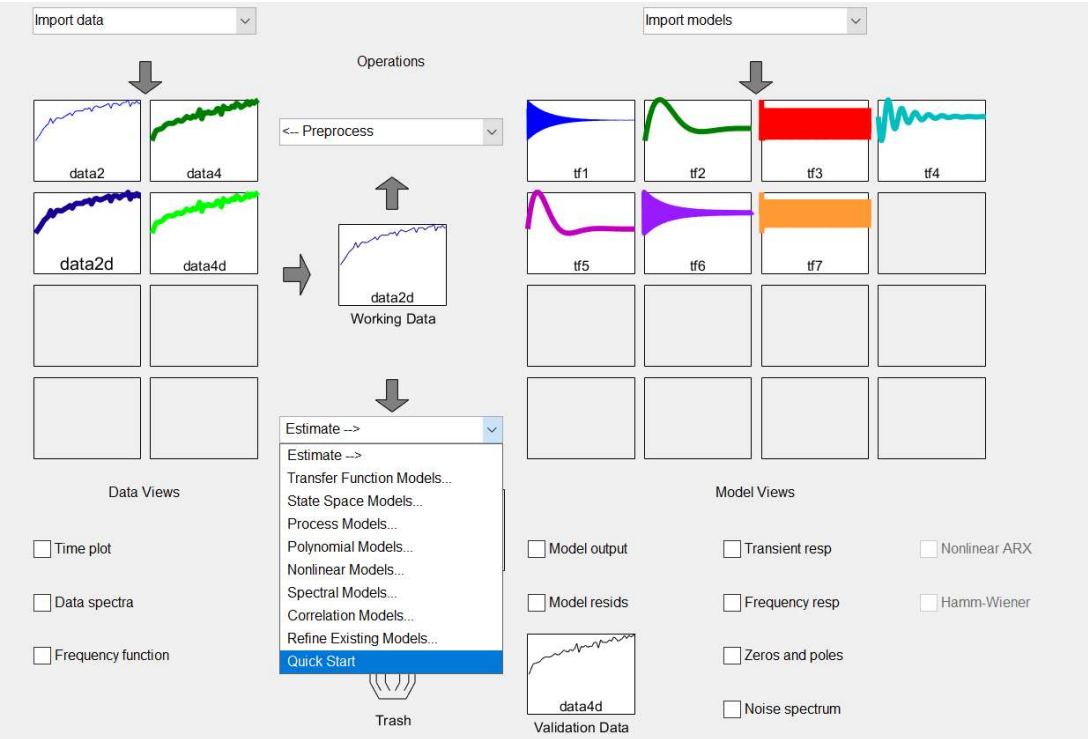

Figure 10. Before Quick Start

In Figure 10, the 2nd data is used as the working data while the 4th data is used as validation data. Then we can validate the Quick Start Models.

4. Validating the Quick Start Models

Quick Start uses three plots, including step-response plot, frequency-response plot and model-output plot, to evaluate the model. This research only discusses the result from model-output plot.

According to the website [1], ”The Model Output window shows agreement among the different model structures and the measured output in the validation data.”

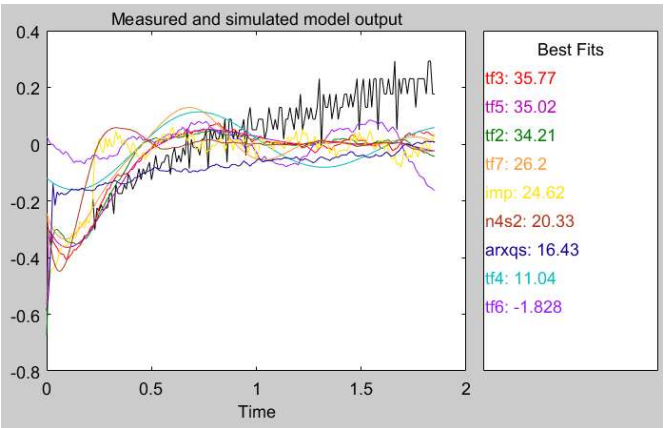

Figure 11. Model-output plot

After setting different number of poles and zeros, the transfer function tf3 in Figure 11 fit our propeller-powered racing vehicle the most.

tf3	tf5	tf2
From input "pwm" to output "velocity": -0.06811 s^3 - 0.6509 s^2 - 12.13 s + 52 ---------- s^4 + 6.329 s^3 + 242.7 s^2 + 1297 s + 7737 Name: tf3 Continuous-time identified transfer function. Parameterization: Number of poles: 4 Number of zeros: 3 Number of free coefficients: 8	From input "pwm" to output "velocity": 163.1 ---------- s^3 + 336.5 s^2 + 2263 s + 1.171e04 Name: tf5 Continuous-time identified transfer function. Parameterization: Number of poles: 3 Number of zeros: 0 Number of free coefficients: 4	From input "pwm" to output "velocity": -0.03217 s^2 - 0.7123 s + 59.5 ---------- s^3 + 62.25 s^2 + 521.4 s + 2943 Name: tf2 Continuous-time identified transfer function. Parameterization: Number of poles: 3 Number of zeros: 2 Number of free coefficients: 6

Table 1. The Best Fits of the transfer function in Figure 11.

5. Results：

By using system identification, the transfer function of the vehicle is modelled as:

$$G(s) = \frac{-0.06811\ ^{3}-0.6509s^2-12.13s+52}{s^4+6.329s^3+242.7s^2+1297s+7737} \tag{1}$$

Also, the model in (1) has a settling time of 12534 seconds and an overshoot of 71.1962 %. To reduce these two values, a controller design is needed.

C. Compensator Design

1. Current Specializations

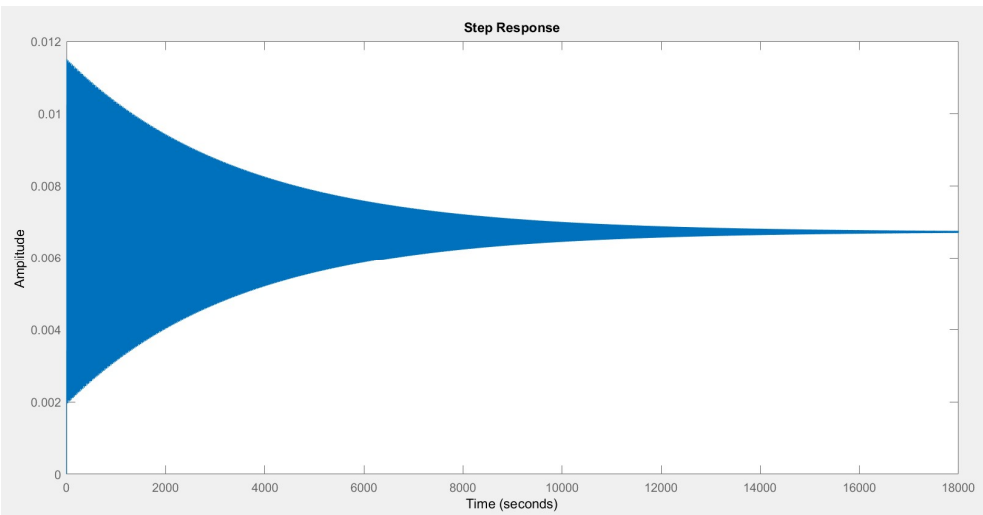

Figure 12. Step response of model (1).

According the model (1) from the system identification, the settling time and the overshoot is too large. We need to add a compensator Gc to improve the performance.

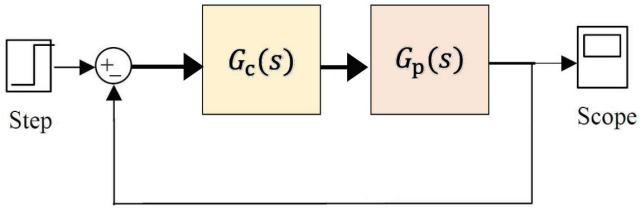

Figure 12. Architecture of Controller Design

Suppose our compensator Gc = $k_1 \frac{s+z}{s+p}$, and Gp is our original model (1). We apply root locus to compensator design by using MATLAB.

2. Procedures

(1) Find all of the poles

(2) Find the dominant pole s_1

(3) Find the compensator angle $\Phi_m = -180° + \sum \theta_{pi} - \sum \theta_{zi}$.

(4) Set the zero position of the compensator and calculate θ_{zc}.

(5) Find the pole location of the compensator, which meets the equation: $\theta_{pc} = \theta_{zc} - \Phi_m$.

(6) Calculate $k_1 = \frac{|s_1-p_1||s_1-p_2||s_1-p_3||s_1-p_4||s_1-p|}{|s_1-z_1||s_1-z_2||s_1-z_3||s_1-z|}$.

We not only target the idea compensator, but also modify the overshoot and the settling time with respect to the changes of poles and zeros.

3. Simulation – Stage 1：

Fixed: Overshoot = 5%

Variable: Settling Time

The resultant compensator is Gc = $k_1 \frac{s+z}{s+p}$，where the values of k_1, z, p are shown below.

fixed overshoot	5%	5%	5%	5%	5%	5%
setting time	0.7 (s)	1.2 (s)	1.7 (s)	2.2 (s)	2.7 (s)	3.2 (s)
k_1	-450.24	-34.82	-61.56	-78.11	-88.74	-96.24
z	-5.71	-3.33	-2.35	-1.82	-1.48	-1.25
p	38.71	-3.91	-3.45	-2.67	-2.12	-1.74
resultant overshoot	NaN	42.8%	36.9%	32.2%	27.9%	26.7%
resultant setting time	NaN	3.70 (s)	2.22 (s)	2.04 (s)	2.45 (s)	2.87 (s)

From the above Table, the overshoot is fixed. When the settling time becomes larger, the absolute value of z $|z|$ becomes smaller, and $|p|$ also becomes smaller.

Besides, although the resultant overshoot is still very large, apparently it becomes smaller compared to the original model.

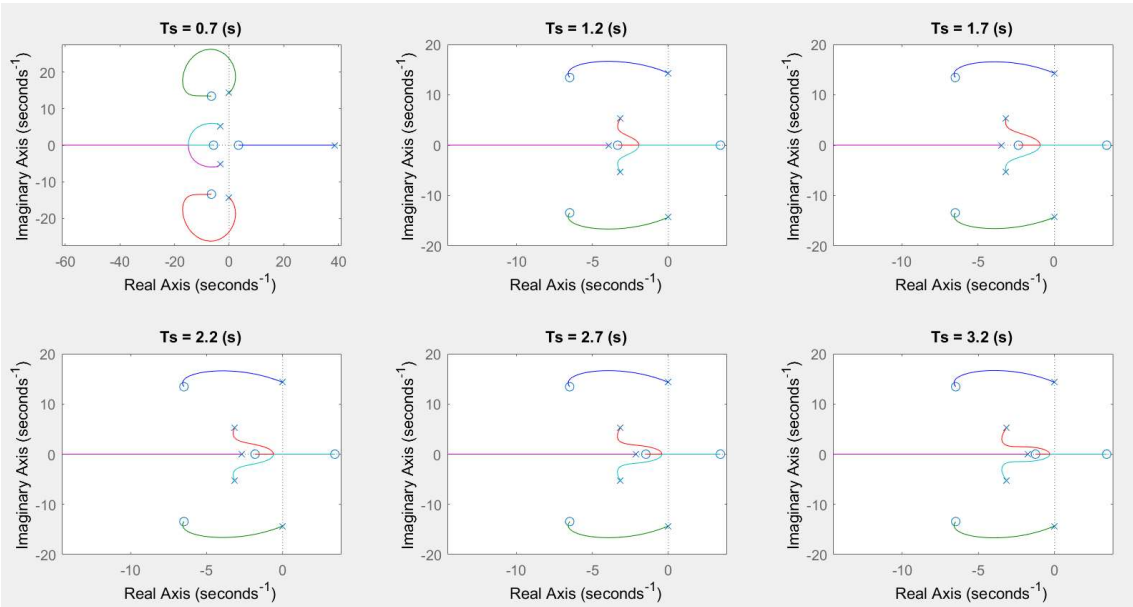

Figure 13. Root locus (fixed overshoot).

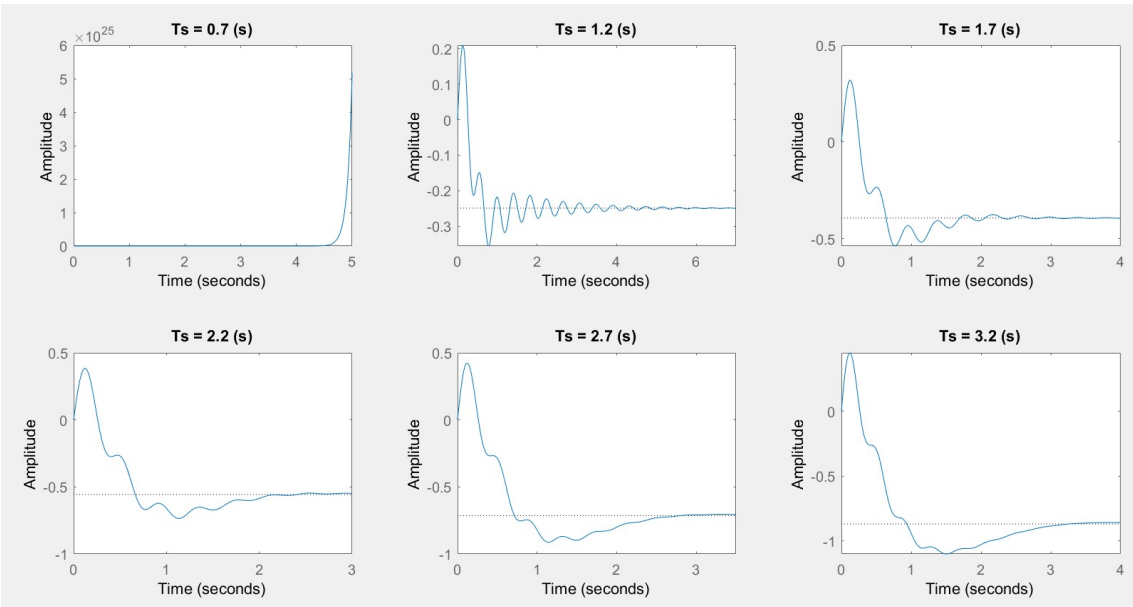

Figure 14. Step Response (fixed overshoot).

The system will diverge is the settling time is smaller than 0.8 second.

If we change the settling time to a larger value, we have the following result:

Fixed overshoot	5%	5%	5%	5%	5%	5%
setting time	2 (s)	3 (s)	4 (s)	5 (s)	6 (s)	7 (s)
k_1	-72.45	-93.51	-104.69	-111.87	-116.96	-120.78
z	-2.00	-1.33	-1.00	-0.80	-0.67	-0.57
p	-2.95	-1.88	-1.34	-1.03	-0.83	-0.69
resultant overshoot	32.72%	26.8%	25.0%	24.4%	24.1%	23.9%
resultant setting time	2.00 (s)	2.73 (s)	3.55 (s)	4.38 (s)	5.21 (s)	6.05 (s)

From the above Table, the overshoot is fixed. When the settling time becomes larger, then $|k_1|$ becomes larger, $|z|$ becomes smaller, and $|p|$ becomes smaller.

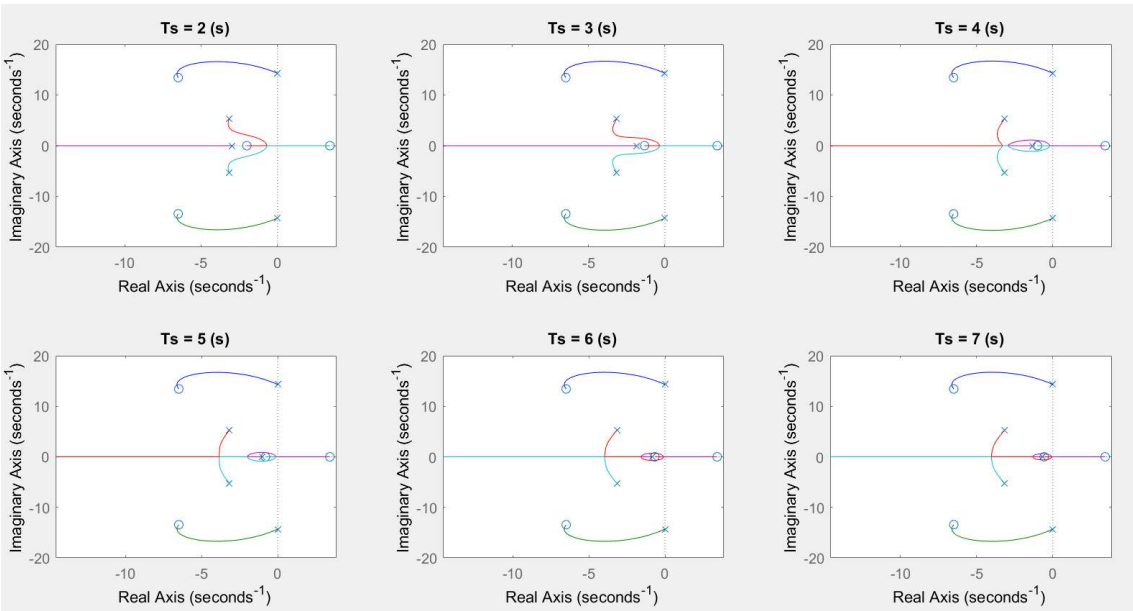

Figure 15. Root locus (fixed overshoot).

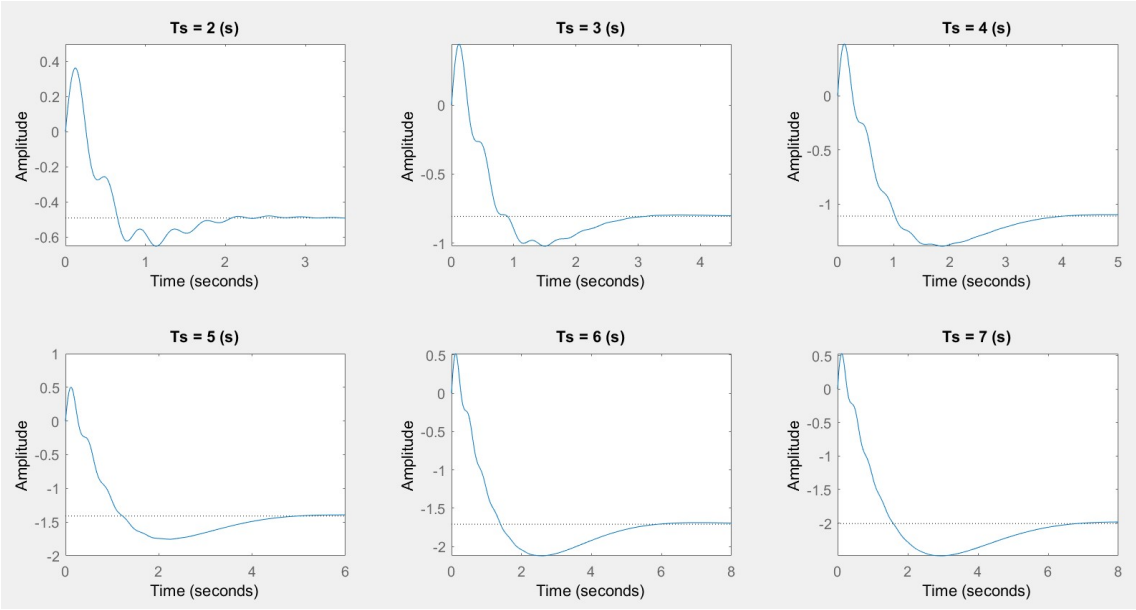

Figure 16. Step Response (fixed overshoot).

4. Simulation – Stage 2：

Fixed: Settling Time = 3 seconds

Variable: Overshoot

The resultant compensator is Gc = $k_1 \frac{s+z}{s+p}$，where the values of k_1, z, p are shown in the next page.

Fixed setting time	3 (s)	3 (s)	3 (s)	3 (s)	3 (s)	3 (s)
overshoot	3%	5%	7%	9%	11%	13%
k_1	-94.03	-93.51	-93.05	-92.63	-92.27	-91.96
z	-1.33	-1.33	-1.33	-1.33	-1.33	-1.33
p	-1.72	-1.88	-2.03	-2.20	-2.38	-2.59
resultant setting time	2.96 (s)	2.73 (s)	2.47 (s)	3.27 (s)	3.25 (s)	3.12 (s)
resultant overshoot	17.8%	26.8%	36.2%	47.4%	58.0%	67.7%

From the above Table, the settling time is fixed. When the overshoot becomes larger, then $|k_1|$ becomes smaller, $|z|$ changes little, and $|p|$ becomes larger.

Besides, the all of the resultant settling time is close to 3 seconds, but the resultant overshoot increases more than the original fixed one.

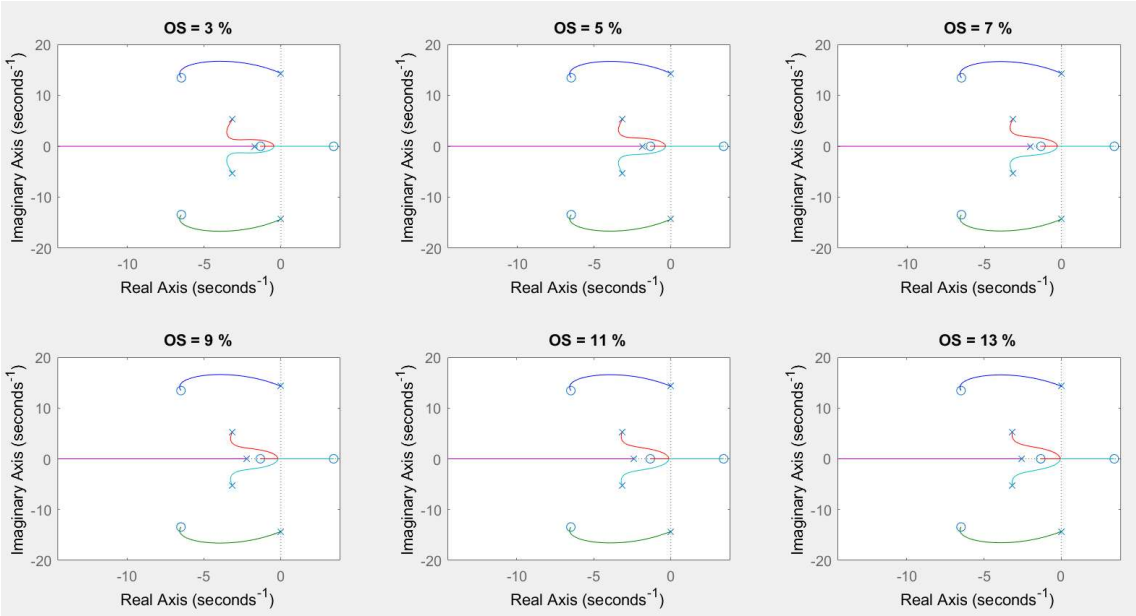

Figure 17. Root locus (fixed settling time)

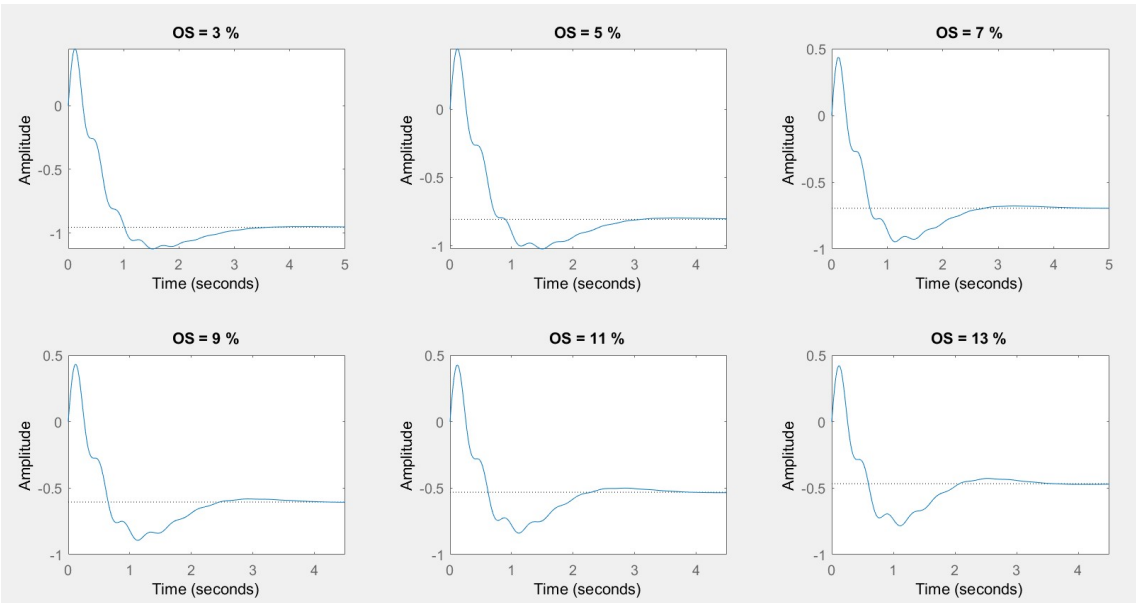

Figure 18. Step Response (fixed settling time)

5. Conclusion

Although the terrible model G_p does not perform well, the new compensator G_c reduces the settling time and the overshoot. The transfer function of the best compensator is:

$$G_c(s) = -94.03\frac{s-1.33}{s-1.72} \tag{2}$$

After adding the compensator (2), the overshoot is reduced to 17.8%, and the settling time is reduced to 2.96 seconds. The compensator finally improves the performance of our propeller-powered racing vehicle.

Note: Our complete 204-page report is on the official website:
http://www.me.ntu.edu.tw/capstone/finalreport_2019/G12.pdf

D. Reference

[1] https://www.mathworks.com/help/ident/gs/identify-linear-models-using-the-gui.html